Assignment

Course Code: CS6007 Instructor: Dr. Avishek Ghosh

Date: August 30, 2025

Instructions:

- Total marks 15
- Please submit your typed solutions(no handwritten).
- Submission Deadline: 9 September(Submit in Class).

Q1. [**Stochastic coordinate model**][1.5+1.5] Suppose we define an L-Lipschitz d-dimensional function $f(x)$ as

$$f(x) = f(x_1, x_2, \dots, x_d).$$

Consider $\xi \in \{1, 2, 3, \dots, d\}$ chosen uniformly at random with $P(\xi = i) = \frac{1}{d}$ for all $i = 1, 2, \dots, d$. We compute

$$G(x;\xi) = d\left[\frac{\partial f(x)}{\partial x_\xi}\right] e_\xi \quad \text{where } e_\xi^\top = [0 \ \dots \ 1 \ \dots \ 0],$$

where the non-zero entry is at the ξ-th location. Show that
(a) $\mathbb{E}_\xi[G(x;\xi)] = \nabla f(x)$ and
(b) $\mathbb{E}_\xi[\|G(x;\xi)\|^2] \leq dL^2$.

Q2.[2+2] Given a nonempty, closed set $\Omega \subset \mathbb{R}^d$, the *distance function* associated with Ω is defined by

$$\operatorname{dist}(x, \Omega) := \inf\{\|x - w\| \mid w \in \Omega\}. \tag{1}$$

Show that:

1. $\operatorname{dist}(\cdot; \Omega)$ is Lipschitz continuous with Lipschitz constant 1.
2. $\operatorname{dist}(\cdot; \Omega)$ is convex if and only if Ω is convex.

Q3. [**Sub-gradients and Optimality**][1.5+2.5]

For a convex function $f : \mathbb{R}^d \to \mathbb{R}$, a subgradient at $x \in \mathbb{R}^d$ is a vector $g \in \mathbb{R}^d$ such that

$$f(y) \geq f(x) + \langle g, y - x\rangle \quad \text{for all}\, y \in \mathbb{R}^d.$$

In this case, we write $g \in \partial f(x)$. We say that f is sub-differentiable when $\partial f(x)$ is non-empty for each $x \in \mathbb{R}^d$.

(a) Show that x^* is a minimizer of f if $0 \in \partial f(x^*)$.

(b) Show that f is Lipschitz with parameter L (i.e., $|f(x) - f(y)| \leq L\|x - y\|_2$ for all $x, y \in \mathbb{R}^d$) if and only if $\|g\|_2 \leq L$ for all subgradient vectors g.

Q4.[**(Contractive property for gradient operators)**][4]

Let $S(x) = x - \alpha \nabla f(x)$ where f is (M, m) smooth and strongly convex with $\alpha = \frac{2}{M+m}$. Show that

$$\|S(x) - S(y)\| \leq \left(\frac{1 - \frac{m}{M}}{1 + \frac{m}{M}} \right) \|x - y\|$$

As a corollary of this show that if GD is run on (M, m) smooth and strongly convex function with $\alpha = \frac{2}{m+M}$, we have

$$\|x^k - x^*\| \leq \left(\frac{1 - \frac{m}{M}}{1 + \frac{m}{M}} \right)^k \|x^0 - x^*\| \quad ; \quad k = 1, 2, \ldots$$

End of Assignment